Diffusion Models

How AI Learns to Create Images from Noise

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CSE 200: Technical Writing and Presentation
Bangladesh University of Engineering and Technology

Outline

Introduction

Training

Applications

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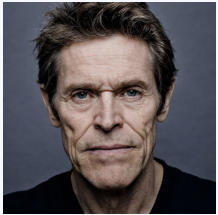
What Problem Are We Solving?

Real face



AI-generated face *(Sora)*

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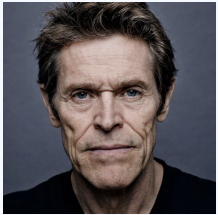

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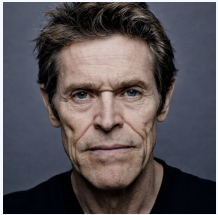

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Diffusion models provide a *stable*, *high-quality* solution.

The Core Idea: Learning by Destroying

Clean Image

contains
all the detail

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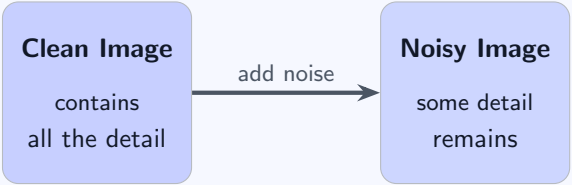

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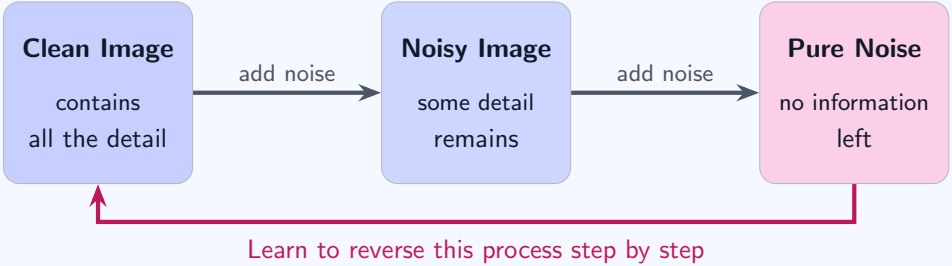

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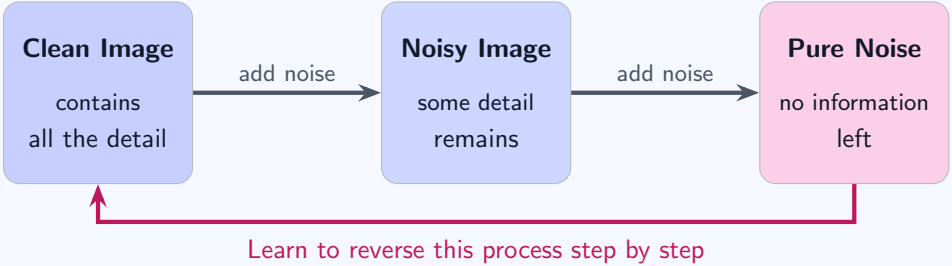

We Essentially Have Two Diffusion Processes

- The **Forward Diffusion Process** that gradually adds noise.
- The **Reverse Diffusion Process** that removes noise step by step.

The Forward Diffusion Process: Adding Noise

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Slightly Noisy

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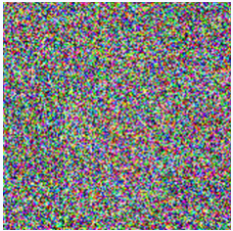

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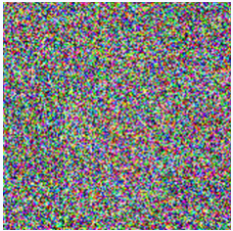
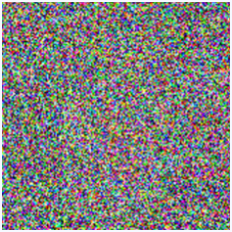


Pure Noise

What Happens

Take a real image and add a *tiny* bit of random noise. Repeat this hundreds of times until only noise remains.

The Reverse Diffusion Process: Removing Noise



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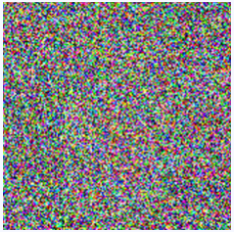

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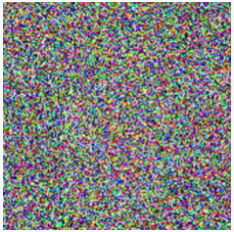

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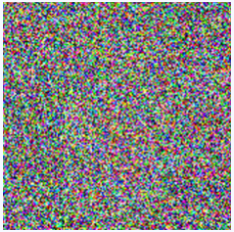

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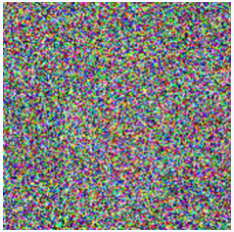

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How Generation Works

- Start from **pure noise**
- Predict and remove a little noise at each step
- Repeat until a realistic image appears

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But how does the model learn this reverse process?

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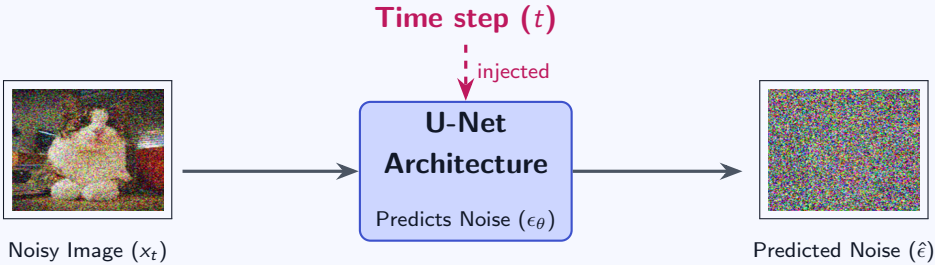
4. Compare the prediction to the **actual noise**, minimize the error

The "Brain" of the Model: The U-Net



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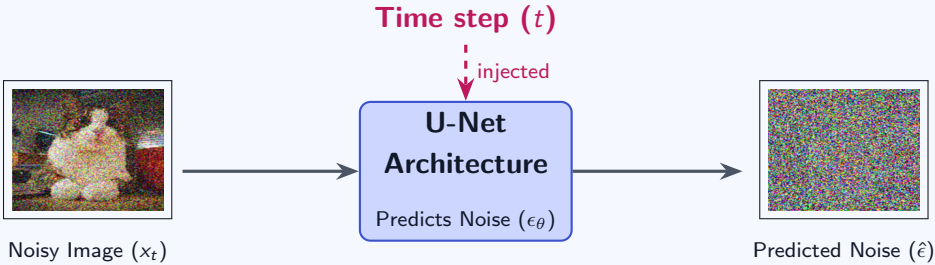

- **Input:** A noisy image and the specific time step (t).
- **Process:** The U-Net shrinks the image to understand the "global context," then expands it back, using **skip connections** to retain fine, pixel-level details.
- **Output:** An exact map of the noise predicted to have been added at step t.

Key Insight & Why It Works

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Why It Works

- The task is simplified into **small, predictable steps**
- Each step only removes a tiny amount of noise
- Repeating simple steps reconstructs complex structure

Why Diffusion Models Work So Well

Advantages

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In practice, diffusion models combine reliability with strong output quality.

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But where is Diffusion Models actually applied?

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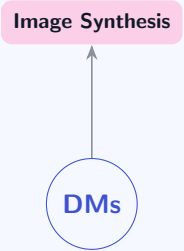

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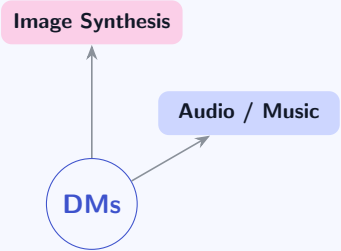

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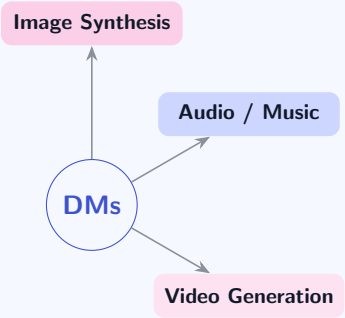

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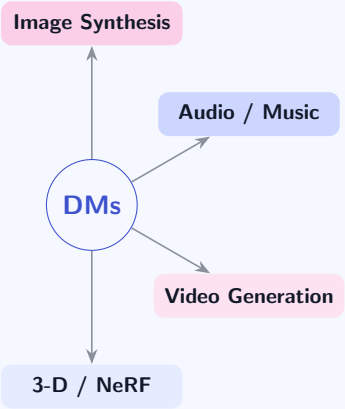

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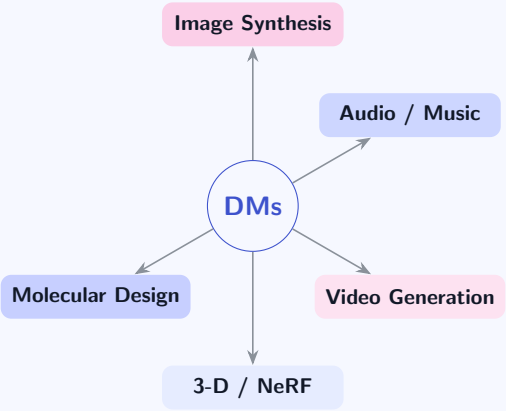

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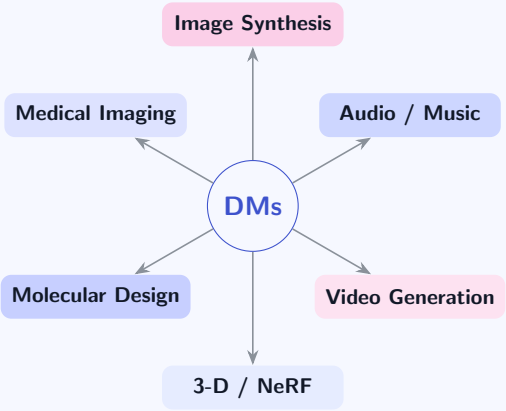

Some Real-World Tools Powered by Diffusion

DALL·E

Text to Image (DALL·E)

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Artistic Image Synthesis (Midjourney)

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Text to Video (Sora)

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- **High compute cost:** training requires significant GPU resources
- **Large model size:** not easy to run on consumer hardware

Open Challenges

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- How do we ensure generated content is **safe and ethical**?
- Can we extend to **longer videos** and **interactive content**?

References

[1] J. Ho et al. **"Denoising diffusion probabilistic models".** In: *Proceedings of the 34th International Conference on Neural Information Processing Systems*. NIPS '20. Vancouver, BC, Canada: Curran Associates Inc., 2020.

[2] Y. Song and S. Ermon. **"Generative modeling by estimating gradients of the data distribution".** In: *Advances in neural information processing systems* 32 (2019).

[3] J. Song et al. **"Denoising diffusion implicit models".** In: *arXiv preprint arXiv:2010.02502* (2020).

[4] A. Q. Nichol and P. Dhariwal. **"Improved denoising diffusion probabilistic models".** In: *International conference on machine learning*. PMLR. 2021, pp. 8162–8171.

Thank You!

Thanks for listening!